BRIGHTLEARN TUTORIALS

DATA SCIENCE

SQL FUNDAMENTALS

EXERCISE 3 : SQL CASE Statements

1. Table : products

SQL query to classify product by price

Syntax :

```
SELECT product_name, price
CASE
    WHEN price >1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000 THEN
    'Mid-range'
    ELSE 'Budget'
END AS price_category
FROM products;
```

Output :

product_name	price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

2. Table : orders

SQL query to label value of orders

Syntax :

```
SELECT customer_name,
       amount,
```

```
    CASE
        WHEN amount >= 1000 THEN 'High_value'
        WHEN amount BETWEEN 500 AND
         999.99 THEN 'Medium_Value'
        ELSE 'Low_Value'
    END AS order_value_category
FROM orders;
```

Output :

Customer_name	amount	order_value_category
Alice	150	Low_value
Bob	560	Medium_value
Charlie	999.99	Medium_value
Diana	45.50	Low_value
Ethan	1200	High_value

3. Table : employees

SQL query to categorize employee position

Syntax :

```
SELECT emp_name, department, salary
  CASE
    WHEN department = 'IT' AND salary > 80000
     THEN 'Senior IT'
    WHEN department = 'HR' AND salary > 55000
     THEN 'Experienced HR'
    ELSE 'Staff'
  END AS position_level
FROM employees;
```

Output :

emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Table : students

SQL query to assign a letter grade

Syntax :

```
SELECT student_name, store,
 CASE
     WHEN score >= 90 THEN 'A'
     WHEN score BETWEEN 80 AND 89 THEN 'B'
     WHEN score BETWEEN 70 AND 79 THEN 'C'
     WHEN score BETWEEN 60 AND 69 THEN 'D'
      ELSE 'F'
 END AS grade
FROM students;
```

Output :

Student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. Table : deliveries

SQL query to label delivery performance

Syntax :

```
SELECT delivery_id, delivery_time_minutes,
CASE
  WHEN delivery_time_minutes <= 30 THEN 'Fast'
  WHEN delivery_time_minutes BETWEEN 31 AND
   60 THEN 'on Time'
  ELSE 'Late'
END AS performance
FROM deliveries;
```

Output :

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6. Table : tickets

SQL query to convert priority to labels

Syntax :

```
SELECT issue_type, priority,
 CASE
   WHEN priority = 3 THEN 'High'
   WHEN priority = 2 THEN 'Medium'
   WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM tickets;
```

Output :

Issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. Table : attendance

SQL query to calculate attendance % and classify

Syntax :

```
SELECT Student_id,
(days_present *100/total_days) AS
      percentage_ attendance,
CASE
 WHEN percentage_ attendance >= 90 THEN
  'Excellent'
 WHEN percentage_attendance BETWEEN
  75 AND 89 THEN 'Good'
 ELSE 'Needs Improvement'
 END AS attendance_status
FROM attendance;
```

Output :

Student_id	percentage_attendance	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

8. Table : products_Inventory
SQL query to label stock status

Syntax :

```
SELECT product_id, stock_qty,
CASE
  WHEN stock_qty = 0 THEN 'Out of stock'
  WHEN stock_qty BETWEEN 1 AND 5 THEN
     'Low stock'
  ELSE 'In stock'
END AS stock_status
FROM product_inventory;
```

Output :

product_id	stock_qty	Stock_Status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

9. Table : classes
SQL query to classify class by size

Syntax :

```
SELECT subject, enrolled_students,
CASE
  WHEN enrolled_Students >= 25 THEN 'Large'
  WHEN enrolled_students BETWEEN 10 AND
     24 THEN 'Medium'
  ELSE 'Small'
```

```
    END AS class_size_category
FROM classes;
```

Output:

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Table : payments

SQL query to apply discount flag

Syntax :

```
SELECT payment_id, payment_method,
       amount,
CASE
  WHEN payment_method = 'cash' AND
    amount >= 200 THEN 'Eligible for Discount'
  ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;
```

Output :

payment_id	payment_method	amount	discount_eligibility
1	Card	50	Not Eligible
2	Cash	200	Eligible for Discount
3	Card	150	Not Eligible
4	Paypal	75	Not Eligible
5	Cash	300	Eligible for Discount